

ALI AKARMA

Trustworthy AI & Agentic Systems Researcher
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RESEARCH PROFILE

Undergraduate AI researcher specializing in trustworthy agentic systems, adversarial robustness, safety-constrained reinforcement learning, and human oversight for autonomous AI. For Fall 2027, I aim to extend this work to trustworthy multimodal and speech-enabled AI, including robust audio-language models, voice agents, and safety evaluation of multimodal agents.

RESEARCH INTERESTS

Trustworthy and safe agentic AI; adversarial robustness and security of LLM-based autonomous systems; safety-constrained multi-agent reinforcement learning; human oversight and value-aligned control; trustworthy multimodal and speech-enabled AI; robust audio-language models and voice agents; multimodal AI safety evaluation.

EDUCATION

B.S. Information Technology — Islamic University of Madinah, Saudi Arabia 2023 – May 2027

- GPA: 4.31 / 5.00 — *Merit-based fully funded scholarship (competitive international selection)*
- Relevant coursework: Machine Learning, Algorithms & Data Structures, Probability & Statistics, Database Systems, Software Engineering
- Thesis interest: Trustworthy multimodal agent orchestration under adversarial conditions

RESEARCH EXPERIENCE

Research Intern — King Fahd University of Petroleum and Minerals (KFUPM) Jun 2026 -- Present

- Investigating verification of autonomous AI decision-making using reinforcement learning and multi-agent systems, with emphasis on safety, reliability, distributional shift, and adversarial conditions.
- Developing evaluation protocols for trustworthy autonomous-agent policies under distribution shift and adversarial scenarios.

Undergraduate AI Researcher — Islamic University of Madinah Jan 2025 -- Present

- Designed constrained multi-agent reinforcement learning architectures using C-Dec-POMDP and Lagrangian CTDE for safety critical prediction, achieving 81.5 reward with a 2.3% safety-violation rate across six baselines.
- Evaluated adversarial prompt-injection and jailbreak failure modes in autonomous LLM-agent pipelines; developed LangChain defence middleware with three detection modes and four policy strategies.
- Developed a governance-constrained agentic AI framework incorporating human oversight for safety-critical autonomous decision-making; first author, ICETAS 2026.
- Built 20+ reproducible open-source research systems with automated tests and Zenodo-archived datasets across agentic AI, safety-critical decision-making, healthcare, and smart infrastructure.

RESEARCH PROJECTS & GRANTS

King Salman Center for Disability Research (KSCDR) — Research Group No. KSRG-2026-321

- ADAPT**: An agentic AI framework for disability-inclusive nutrition and healthcare monitoring. *Published in Journal of Disability Research*.

DOI: [10.57197/JDR-2026-0830](https://doi.org/10.57197/JDR-2026-0830) | GitHub: github.com/aliakarma/ADAPT

- FedAgent-Chain**: A secure federated and agentic AI framework for multilingual disability-inclusive employment in AI cities. *Published in MDPI Smart Cities*.

DOI: [10.3390/smartcities9070106](https://doi.org/10.3390/smartcities9070106) | GitHub: github.com/aliakarma/fedagent-chain

SELECTED RESEARCH PUBLICATIONS

Google Scholar: 71 citations | h-index: 5 | i10-index: 2 (September 2026).

First-Author Papers

- **Akarma, A.**, Syed, T. A., Khan, M., & Ahmad, A. (2026). Privacy Leakage in Federated Learning: Gradient-Based Client Identity Inference and Defenses for Inertial Sensing in Vehicular Edge Networks. **104th VTC Fall Boston** [arXiv:2609.02971](#)
- **Akarma, A.**, Syed, T.A., Jan, S., Muneer, H., Jilani, A.K. "Governance-Constrained Agentic AI: Blockchain-Enforced Human Oversight for Safety-Critical Wildfire Monitoring," **ICETAS 2026**. [arXiv:2604.04265](#)
- **Akarma, A.**, Syed, T.A., Jilani, A.K., Jan, S., Muneer, H., Khan, M.A., Yu, C. "Agents for Agents: An Interrogator-Based Secure Framework for Autonomous Internet of Underwater Things," **ICETAS 2026**. [arXiv:2604.04262](#)
- **Akarma, A.**, Syed, T.A., Hameed, D., Muneer, H. "Improving Disability Support and Institutional Performance With AI," Book Chapter, **IGI Global**. [preprints202608.1878](#)

Journal Articles

- Syed TA, **Akarma A.**, Alatify A, Naqash MT, Alqurashi A. "Agentic AI-enhanced digital twins for Smart City civil infrastructure: A secure, autonomous and auditable management framework" **PLOS ONE**. [10.1371/journal.pone.0353610](#) (Co-First Author).
- Jan, S., **Akarma, A.**, Syed, T.A. et al. "EAGF: a four-pillar ethical AI governance framework for trustworthy cybersecurity in 5G renewable energy IoT systems". **Scientific Reports** 2026. [10.1038/s41598-026-63383-5](#).
- Syed, T. A., **Akarma, A.**, Naqash, M. T., Hameed, D., Kamal, S., & Formisano, A. "Agentic AI for Climate-Resilient Cities: A PRISMA-Guided Review and Digital Twin Framework," **MDPI Sustainability**. [10.3390/su18178917](#)
- Syed, T. A., Alshahrani, A., **Akarma, A.**, Khan, S., Nauman, M., Lee, I. E., ... & Ullah, A. "FinNutriAgent (FNA): An Agentic AI for Budget and Nutrition Planning," **ETASR**, 2026. [10.48084/etasr.15640](#)

Conference Papers

- Syed, T. A., Alshahrani, A., Ullah, A., **Akarma, A.**, Khan, S., Nauman, M., & Jan, S. "FinAgent: Agentic AI for Personal Finance and Nutrition," **IEEE ICCA 2025**. [10.1109/ICCA66035.2025.11430760](#)
- Jan, S., Razzaqi, H. A., **Akarma, A.**, & Belgaum, M. R "Blockchain-Monitored Agentic AI Architecture for Trusted Perception-Reasoning-Action Pipelines," **IEEE ICCA 2025**. [10.1109/ICCA66035.2025.11430865](#)

OPEN-SOURCE RESEARCH SOFTWARE

All repositories include reproducible experimental pipelines and Zenodo-archived datasets.

- **Trustguard** — Constrained multi-agent reinforcement learning framework for trustworthy runtime permission governance using MAPPO and Lagrangian safety constraints. ([Repository](#))
- **Secure Agent Runtime** — Multimodal security framework for autonomous LLM agents, covering prompt injection, provenance, and text, image, audio, and video threats. ([Repository](#))
- **SAS-Shield** — Safety-assurance framework for agentic AI combining conformal testing, formal verification, control-barrier constraints, adversarial evaluation, and cryptographic auditability. ([Repository](#))

Additional research software and reproducibility artifacts: [github.com/aliakarma](#)

TECHNICAL SKILLS

AI/ML: PyTorch, Hugging Face Transformers, scikit-learn, NumPy

Agentic AI & LLMs: LLM orchestration, autonomous agents, prompt-injection defense, adversarial robustness

Reinforcement Learning: MAPPO, CTDE, C-Dec-POMDP, Lagrangian constrained optimization

Trustworthy AI: safety evaluation, failure-mode analysis, human oversight, policy-aligned control, auditability

Research Engineering: Python, Git, Linux, Jupyter, pytest, LaTeX, Zenodo

AWARDS & CERTIFICATIONS

- **Merit-Based Fully Funded Scholarship** — Islamic University of Madinah (competitive international selection, 2023)
- **Certificate of Research Excellence** — ICETAS 2026, for research contribution on agentic AI governance
- **Certificate of Appreciation** — ICBDT 2025, for presented research contributions

LANGUAGES

Languages: Urdu (native), English (professional proficiency), Arabic (professional proficiency)